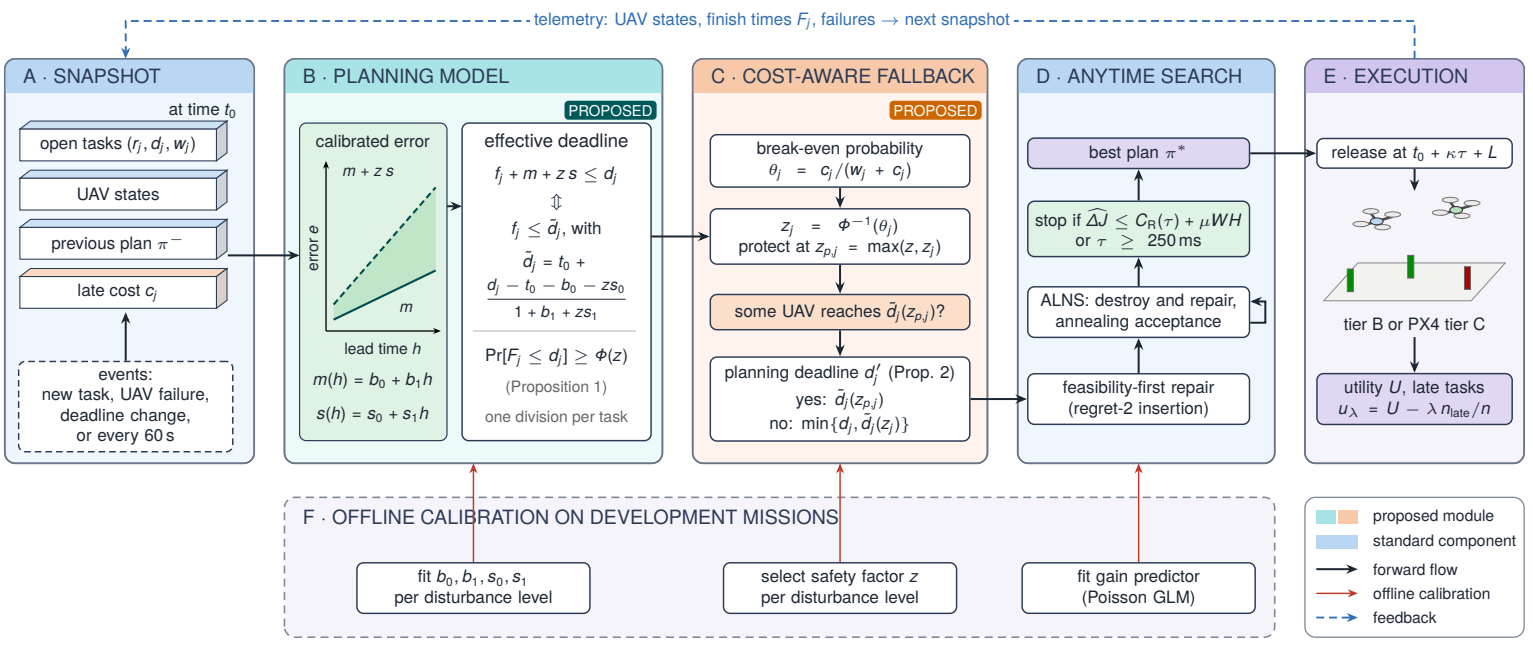

telemetry: UAV states, finish times F_j, failures → next snapshot
A · SNAPSHOT
at time t_0
open tasks (r_j, d_j, w_j)
UAV states
previous plan π^-
late cost c_j
events: new task, UAV failure, deadline change, or every 60 s
B · PLANNING MODEL
PROPOSED
calibrated error
$m + z\,s$
error e
m
lead time h
$m(h) = b_0 + b_1 h$
$s(h) = s_0 + s_1 h$
effective deadline
$f_j + m + z\,s \le d_j$
⇕
$f_j \le \tilde{d}_j$, with
$\tilde{d}_j = t_0 + \dfrac{d_j - t_0 - b_0 - z s_0}{1 + b_1 + z s_1}$
$\Pr[F_j \le d_j] \ge \Phi(z)$
(Proposition 1)
one division per task
C · COST-AWARE FALLBACK
PROPOSED
break-even probability
$\theta_j = c_j / (w_j + c_j)$
$z_j = \Phi^{-1}(\theta_j)$
protect at $z_{p,j} = \max(z, z_j)$
some UAV reaches $\tilde{d}_j(z_{p,j})$?
planning deadline d'_j (Prop. 2)
yes: $\tilde{d}_j(z_{p,j})$
no: $\min\{d_j, \tilde{d}_j(z_j)\}$
D · ANYTIME SEARCH
best plan π^*
stop if $\widehat{\Delta J} \le C_R(\tau) + \mu WH$ or $\tau \ge 250$ ms
ALNS: destroy and repair, annealing acceptance
feasibility-first repair (regret-2 insertion)
E · EXECUTION
release at $t_0 + \kappa\tau + L$
tier B or PX4 tier C
utility U, late tasks
$u_\lambda = U - \lambda\, n_{\text{late}}/n$
F · OFFLINE CALIBRATION ON DEVELOPMENT MISSIONS
fit b_0, b_1, s_0, s_1 per disturbance level
select safety factor z per disturbance level
fit gain predictor (Poisson GLM)
proposed module
standard component
forward flow
offline calibration
feedback